HILB^D OF A SCHEME WITH AFFINE-OPENS THROUGH POINTS IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 515-522 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 524-586, the complete author proof (61 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_pic.json (pb=3101, stmt 515-522, proof 524-586).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the details of the reduction to affine S (two standard alternative routes named) - a named routine reduction; the F_U gluing argument including the non-separated footnote is in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 01LG: sources/stacks-pic/pinned-original-source.tex, source lines 586–586. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B99: sources/stacks-pic/pinned-original-source.tex, source lines 411–414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B95: sources/stacks-pic/pinned-original-source.tex, source lines 282–286. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 01W6: sources/stacks-pic/pinned-original-source.tex, source lines 586–586. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 01JJ: sources/stacks-pic/pinned-original-source.tex, source lines 586–586. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 01IQ: sources/stacks-pic/pinned-original-source.tex, source lines 586–586. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let $X \to S$ be a morphism of schemes. Let $d \geq 0$. Assume for all $(s, x_1, \ldots, x_d)$ where $s \in S$ and $x_1, \ldots, x_d \in X_s$ there exists an affine open $U \subset X$ with $x_1, \ldots, x_d \in U$. Then $\mathrm{Hilb}^d_{X/S}$ is representable by a scheme.*

Proof. Either using relative glueing (Constructions, Section 01LG) or using the functorial point of view (Schemes, Lemma 01JJ) we reduce to the case where S is affine. Details omitted.

Assume S is affine. For $U \subset X$ affine open, denote $F_U \subset \mathrm{Hilb}^d_{X/S}$ the subfunctor such that for a scheme T/S an element $Z \in \mathrm{Hilb}^d_{X/S}(T)$ is in $F_U(T)$ if and only if $Z \subset U_T$. We will use Schemes, Lemma 01JJ and the subfunctors F_U to conclude.

Condition (1) is Lemma 0B95.

Condition (2)(a) follows from the fact that $F_U = \mathrm{Hilb}^d_{U/S}$ and that this is representable by Lemma 0B99. Namely, if $Z \in F_U(T)$, then Z can be viewed as a closed subscheme of U_T which is finite locally free of degree d over T and hence $Z \in \mathrm{Hilb}^d_{U/S}(T)$. Conversely, if $Z \in \mathrm{Hilb}^d_{U/S}(T)$ then $Z \to U_T \to X_T$ is a closed immersion[1] and we may view Z as an element of $F_U(T)$.

Let $Z \in \mathrm{Hilb}^d_{X/S}(T)$ for some scheme T over S. Let

$$B = (Z \to T)\big((Z \to X_T \to X)^{-1}(X \setminus U)\big)$$

This is a closed subset of T and it is clear that over the open $T_{Z,U} = T \setminus B$ the restriction $Z_{t'}$ maps into $U_{T'}$. On the other hand, for any $b \in B$ the fibre Z_b does not map into U. Thus we see that given a morphism $T' \to T$ we have $Z_{T'} \in F_U(T')$ $\Leftrightarrow$ $T' \to T$ factors through the open $T_{Z,U}$. This proves condition (2)(b).

Condition (2)(c) follows from our assumption on X/S. All we have to do is show the following: If T is the spectrum of a field and $Z \subset X_T$ is a closed subscheme, finite flat of degree d over T, then $Z \to X_T \to X$ factors through an affine open U of X. This is clear because Z will have at most d points and these will all map into the fibre of X over the image point of $T \to S$. □

[1]This is clear if $X \to S$ is separated as in this case Morphisms, Lemma 01W6 tells us that the immersion $\varphi : Z \to X_T$ has closed image and hence is a closed immersion by Schemes, Lemma 01IQ. We suggest the reader skip the rest of this footnote as we don't know of any instance where the assumptions on $X \to S$ hold but $X \to S$ is not separated. In the general case, let $x \in X_T$ be a point in the closure of $\varphi(Z)$. We have to show that $x \in \varphi(Z)$. Let $t \in T$ be the image of x. By assumption on $X \to S$ we can choose an affine open $W \subset X_T$ containing x and $\varphi(Z_t)$. Then $\varphi^{-1}(W)$ is an open containing the whole fibre Z_t and since $Z \to T$ is closed, we may after replacing T by an open neighbourhood of t assume that $Z = \varphi^{-1}(W)$. Then $\varphi(Z) \subset W$ is closed by the separated case (as $W \to T$ is separated) and we conclude $x \in \varphi(Z)$.